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When Conformal Fusion Cannot Be Calibrated: Attainability Limits for Trace-Based Detection of Prompt-Injection Compromise

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Abstract

Runtime monitors for tool-calling language-model agents merge several trace-derived signals into one alarm, and how they are merged is treated as an implementation detail. We show it is the binding constraint, and that the constraint is a sample-size law. Conformal tail surprisals make incommensurable scales comparable, after which any standard rule can merge them. Our main result is that a conformal p-value from n calibration values never falls below $1/(n+1)$, capping the evidence any e-value calibrator can express at e^{u-1}/u , $u = \log(n+1)$. Since averaging is the only merging rule valid under arbitrary dependence, the dependence-robust level- α test has an *empty* rejection region unless $e^{u-1}/u \geq 1/\alpha$ —inert below $n = 312$ trajectories at $\alpha = 0.05$, while Bonferroni-style rules turn on at $n \geq K/\alpha - 1$. No published prompt-injection corpus reaches the first threshold. On **90** compromised trajectories from two commercial agents, eight rules give pooled AUROC within **0.010** of each other while their a-priori true-positive rates span **0.000** to **0.867**. The e-average, mean- p and the union bound are inert as predicted; Bonferroni alone fires, with power equal to the fraction of positives having one signal at the p-value floor (**0.416**), because both are never there together (**0.000**). Injection evidence is dense in the bulk and sparse in the tail, and those facts select opposite rules. The same budget argument explains why an anytime-valid detector lost to a hand-picked quantile. We give the corpus-design requirement that follows, and release all code and data.

Body text.